

Medical Manager Summary: Latent Causal Modeling for MIMIC-III and PhysioNet 2012 Challenge

Dataset explanation: MIMIC-III is a large ICU electronic health record dataset containing deidentified patient admissions, demographics, ICU stays, vital signs, laboratory tests, medications, procedures, diagnoses, notes, and mortality outcomes. PhysioNet 2012 Challenge is a more compact ICU mortality dataset containing adult ICU stays from cardiac, cardiac surgery recovery, medical, and surgical ICUs, with six admission descriptors and time-stamped physiologic variables collected during the first 48 hours after ICU admission. Both datasets come from real clinical care and therefore contain irregular measurements: some patients are monitored frequently, while others have sparse labs or vitals. Missing data and measurement frequency may reflect clinical concern, severity, unit workflow, or care intensity rather than random absence.

Research method: The proposed approach does not treat raw labs or vitals as the main causal variables. Instead, it models hidden clinical states such as shock, respiratory failure, renal dysfunction, metabolic derangement, inflammation or sepsis burden, neurologic dysfunction, cardiac strain, chronic burden, and global severity. Observed variables such as blood pressure, lactate, creatinine, oxygen saturation, GCS, troponin, urine output, and ventilation status are interpreted as noisy indicators of these latent states. Rule-based decision trees then convert observed data into binary proxy labels such as LAT_SHOCK or LAT_RENAL_DYSFUNCTION.

The key methodological point is that these rule-based trees must be clinically validated and ideally co-constructed by physicians. Thresholds, variable combinations, and missingness rules can strongly affect whether the resulting labels reflect true patient physiology or misleading artifacts of documentation, treatment, or measurement patterns. MIMIC supports richer treatment-aware rules because it includes medications, procedures, diagnoses, and itemized chart events. PhysioNet 2012 supports a narrower but cleaner first-48-hour physiologic implementation; labels such as sepsis, shock, cardiac strain, and chronic burden should be interpreted as proxy states rather than confirmed diagnoses when only PhysioNet variables are available.

Strengths	Weaknesses / Risks
Clinically interpretable latent states instead of black-box raw-variable correlations	Latent states are not directly observed and may be misclassified in both datasets
Aligns with medical reasoning: background risk -> organ dysfunction -> mortality	Rule thresholds may be inaccurate without expert clinical review
Handles ICU complexity better than simple lab-to-outcome models	Missingness is informative and may bias labels, especially lactate, ABG, troponin, urine output, and GCS
Separates physiology from treatments and measurement processes	Treatments such as vasopressors or ventilation can be both severity markers and care decisions; PhysioNet includes ventilation but lacks vasopressor and dialysis data
Produces implementable binary tags for downstream modeling in MIMIC and PhysioNet	ICD codes, charting, medications, procedures, and labs may be incomplete or inconsistently recorded in MIMIC; many of these fields are absent in PhysioNet
Supports causal thinking about confounders, mediators, colliders, and measurement intensity	A DAG is a clinical hypothesis, not proven ground truth
Allows dataset-specific fallbacks while preserving a common latent ontology	The same output label can have different certainty across MIMIC and PhysioNet because variable support differs

Main conclusion. Doctors are required for both the construction and validation of the rule-based decision trees. Without clinical expert input, the model may confuse measurement artifacts, treatment decisions, documentation patterns, or dataset-specific variable availability with true disease states. Physician review is essential to confirm that each latent label represents a medically meaningful condition and that downstream conclusions about mortality, risk, or treatment effects are clinically reliable across both MIMIC and PhysioNet 2012.

Appendix: Variables, Metadata, and Latent-Variable Rules

Scope note: MIMIC-III contains thousands of ITEMID-coded chart, medication, procedure, diagnosis, and lab variables. PhysioNet 2012 Challenge contains a fixed set of general descriptors and time-series variables collected during the first 48 ICU hours. This appendix lists the observed variables required for the proposed latent-variable model and decision-tree implementation, with dataset-specific mappings and fallbacks. Exact MIMIC ITEMIDs should be mapped from D_ITEMS and D_LABITEMS. PhysioNet 2012 variable names should be used as provided in the challenge files. No rule uses mortality, survival time, discharge disposition, hospital length of stay, SAPS-I, SOFA, or post-outcome information.

A. Metadata and Background Variables

Variable	Type	Meaning	Use in model
SUBJECT_ID	Identifier	MIMIC patient identifier	Linking only; not available in PhysioNet
HADM_ID	Identifier	MIMIC hospital admission identifier	Linking only; not available in PhysioNet
ICUSTAY_ID	Identifier	MIMIC ICU stay identifier	Main MIMIC row identifier
RecordID	Identifier	PhysioNet ICU stay identifier	Main PhysioNet row identifier; linking only
AGE / Age	Metadata / background	Age at admission or ICU stay start	Chronic burden, confounding; available in both datasets
GENDER / Gender	Metadata / background	Recorded sex or gender	Background risk; available in both datasets with dataset-specific coding
ETHNICITY	Metadata / background	Recorded ethnicity	MIMIC social and case-mix context; absent in PhysioNet
INSURANCE	Metadata / background	Insurance category	MIMIC social and care-context proxy; absent in PhysioNet
LANGUAGE	Metadata / background	Recorded language	MIMIC social and care-context proxy; absent in PhysioNet
MARITAL_STATUS	Metadata / background	Marital status	Optional MIMIC social-context proxy; absent in PhysioNet
ADMISSION_TYPE	Metadata / background	Emergency, urgent, elective, newborn	MIMIC case-mix and acute-risk proxy; absent in PhysioNet
ADMISSION_LOCATION	Metadata / background	Source of hospital admission	MIMIC case-mix proxy; absent in PhysioNet
FIRST_CAREUNIT	Metadata / care context	First ICU unit	MIMIC ICU type and measurement context
LAST_CAREUNIT	Metadata / care context	Last ICU unit	Care trajectory; usually not used for early labels
ICUType	Metadata / care context	PhysioNet ICU type: CCU, CSRU, MICU, or SICU	PhysioNet case-mix and measurement context; MIMIC analog is FIRST_CAREUNIT
Height	Metadata / background	Patient height in cm	PhysioNet descriptor; can support BMI/chronic burden when paired with weight
Weight	Metadata / background and time series	Admission and serial weight in kg	PhysioNet descriptor and time series; MIMIC has itemized weight charting
INTIME	Metadata / time	MIMIC ICU admission time	Defines MIMIC time window
OUTTIME	Metadata / time	MIMIC ICU discharge time	Do not use for early latent labels
ADMITTIME	Metadata / time	Hospital admission time	Timing only
DISCHTIME	Outcome-adjacent metadata	Hospital discharge time	Avoid in latent labels
DIAGNOSIS	Metadata / text	MIMIC admission diagnosis text	Optional phenotype context; absent in PhysioNet
ICD-9 diagnosis codes	Metadata / phenotype	Diagnosed conditions	MIMIC chronic burden and comorbidity; absent in PhysioNet
ICD-derived comorbidity score	Derived metadata	Charlson or Elixhauser-style burden	MIMIC chronic burden; not available in PhysioNet
HOSPITAL_EXPIRE_FLAG / EXPIRE_FLAG / DEATHTIME / DOD	Outcome	MIMIC mortality outcomes and dates	Outcome only, never used in latent rules
SAPS-I, SOFA, Length of stay, Survival, In-hospital death	Outcome-related descriptors	PhysioNet outcome file variables	Outcome/validation only; never used in latent rules

B. Observed Clinical Variables

B1. Vital Signs and Bedside Charted Variables

Observed variable	Unit	Main latent parent or role	Dataset availability / mapping
Heart rate / HR	beats per minute	Shock, inflammation, cardiac strain, global severity	MIMIC chart events; PhysioNet HR
Systolic blood pressure / SysABP / NISysABP	mmHg	Shock, global severity	MIMIC chart events; PhysioNet invasive and non-invasive variants
Diastolic blood pressure / DiasABP / NIDiasABP	mmHg	Shock, global severity	MIMIC chart events; PhysioNet invasive and non-invasive variants
Mean arterial pressure / MAP / NIMAP	mmHg	Shock, global severity	MIMIC chart events; PhysioNet invasive and non-invasive variants
Respiratory rate / RespRate	breaths per minute	Respiratory failure, metabolic derangement, inflammation	Available in both datasets
Oxygen saturation / SpO2 / SaO2	percent	Respiratory failure	MIMIC SpO2/SaO2 item mapping; PhysioNet SaO2
Temperature / Temp	degrees C after conversion	Inflammation or sepsis burden	Available in both datasets; normalize MIMIC units
Glasgow Coma Scale total score / GCS	3 to 15	Neurologic dysfunction, global severity	Available in both datasets
GCS eye, verbal, motor components	component scores	Neurologic dysfunction	MIMIC only; PhysioNet has total GCS only
RASS score	integer scale	Neurologic dysfunction and sedation context	MIMIC only
Pupil response	categorical	Neurologic dysfunction	MIMIC only
Mechanical ventilation status / MechVent	binary or categorical	Respiratory failure and treatment process	Available in both; PhysioNet MechVent is binary
FiO2	fraction or percent normalized to fraction	Respiratory failure and oxygen support	Available in both; normalize MIMIC percent/fraction representations
PEEP	cmH2O	Respiratory failure and ventilatory support	MIMIC only; absent in PhysioNet
Ventilator mode	categorical	Respiratory treatment process	MIMIC only; absent in PhysioNet

Observed variable	Unit	Main latent parent or role	Dataset availability / mapping
Urine output / Urine	mL or mL per kg per hour	Renal dysfunction and shock	Available in both; PhysioNet Urine is time-series mL
Fluid input	mL	Treatment and volume management	MIMIC only; absent in PhysioNet
Fluid output	mL	Renal function, volume status, care process	MIMIC output events; PhysioNet only has Urine
Serial weight	kg	Volume-status context, weak proxy	MIMIC itemized charting; PhysioNet Weight time series

B2. Laboratory Variables

Observed variable	Unit	Main latent parent or role	Dataset availability / mapping
White blood cell count / WBC	K per uL or cells/nL	Inflammation or sepsis burden	Available in both; normalize units
Hemoglobin	g per dL	Hematologic status, nonspecific severity	MIMIC only; PhysioNet uses HCT instead
Hematocrit / HCT	percent	Hematologic status and fluid context	Available in both; PhysioNet HCT
Platelets	K per uL or cells/nL	Hepatic or coagulation dysfunction	Available in both; normalize units
Sodium / Na	mmol per L or mEq/L	Metabolic derangement	Available in both
Potassium / K	mmol per L or mEq/L	Renal dysfunction and metabolic derangement	Available in both
Chloride	mmol per L	Metabolic context	MIMIC only; absent in PhysioNet
Bicarbonate / HCO3	mmol per L	Metabolic derangement and renal dysfunction	Available in both
Blood urea nitrogen / BUN	mg per dL	Renal dysfunction	Available in both
Creatinine	mg per dL	Renal dysfunction	Available in both
Glucose	mg per dL	Metabolic derangement	Available in both
Lactate	mmol per L	Shock, sepsis burden, metabolic derangement	Available in both; missingness is informative
pH	pH units	Respiratory failure and metabolic derangement	Available in both
PaO2	mmHg	Respiratory failure	Available in both
PaCO2	mmHg	Respiratory failure and acid-base status	Available in both
Base excess	mmol per L	Metabolic derangement and shock	MIMIC only; absent in PhysioNet
Bilirubin	mg per dL	Hepatic dysfunction	Available in both
AST	U per L or IU/L	Hepatic injury	Available in both
ALT	U per L or IU/L	Hepatic injury	Available in both
Alkaline phosphatase / ALP	U per L or IU/L	Hepatic or cholestatic injury	Available in PhysioNet and commonly mappable in MIMIC
Albumin	g per dL	Chronic burden, hepatic dysfunction, nutrition or inflammation proxy	Available in both
INR	ratio	Coagulation dysfunction	MIMIC only; absent in PhysioNet
PT	seconds	Coagulation dysfunction	MIMIC only; absent in PhysioNet
PTT	seconds	Coagulation dysfunction	MIMIC only; absent in PhysioNet
Troponin / Tropl / TropT	assay-specific	Cardiac strain or myocardial injury	Available in both; PhysioNet has Tropl and TropT
CK-MB	assay-specific	Cardiac strain or myocardial injury	MIMIC only; absent in PhysioNet
Magnesium / Mg	mmol per L	Metabolic derangement	Available in both
Cholesterol	mg per dL	Weak chronic or nutritional proxy	PhysioNet only; usually not central to latent labels

B3. Treatment and Care-Process Variables

Variable	Type	Main role	Dataset availability / mapping
Vasopressor use	Treatment	Treatment response to shock	MIMIC available through medication/input events; absent in PhysioNet
Norepinephrine use	Treatment	Shock treatment proxy	MIMIC only
Epinephrine use	Treatment	Shock treatment proxy	MIMIC only
Dopamine use	Treatment	Shock treatment proxy	MIMIC only
Phenylephrine use	Treatment	Shock treatment proxy	MIMIC only
Vasopressin use	Treatment	Shock treatment proxy	MIMIC only
Dobutamine use	Treatment	Cardiac support or shock treatment proxy	MIMIC only
Mechanical ventilation / MechVent	Treatment	Respiratory support proxy	Available in both; in PhysioNet use MechVent binary variable
Noninvasive ventilation	Treatment	Respiratory support proxy	MIMIC only; absent in PhysioNet
Dialysis or CRRT	Treatment	Severe renal dysfunction or metabolic failure proxy	MIMIC only; absent in PhysioNet
Antibiotic administration	Treatment	Suspected infection proxy	MIMIC only; absent in PhysioNet
Culture ordered	Care-process variable	Infection suspicion and measurement process	MIMIC only; absent in PhysioNet
Positive culture	Observed clinical result	Infection evidence, imperfect proxy	MIMIC only; absent in PhysioNet
Lab measurement count	Measurement process	Observation intensity	Both datasets: use MIMIC event counts or PhysioNet time-series counts
Vital-sign measurement count	Measurement process	Observation intensity	Both datasets: use MIMIC chart-event counts or PhysioNet time-series counts
Missingness indicator per variable	Measurement process	Informative missingness	Available in both after preprocessing

C. Final Latent Variables

Latent variable	Meaning	Output column
Chronic burden	Baseline vulnerability and comorbidity burden; MIMIC can use ICD/comorbidity data, while PhysioNet uses age, ICUType, BMI, albumin, and body-size proxies	LAT_CHRONIC_BURDEN
Inflammation or sepsis burden	Systemic inflammation or suspected infection; confirmed suspected infection is stronger in MIMIC, while PhysioNet supports inflammation/sepsis-like burden only	LAT_INFLAMMATION_SEPSIS
Global severity	Multi-domain acute physiological severity across hemodynamic, respiratory, neurologic, renal, metabolic, inflammatory, hepatic/coagulation, and cardiac domains	LAT_GLOBAL_SEVERITY
Shock	Circulatory failure or tissue hypoperfusion; MIMIC can add vasopressors, PhysioNet uses BP, lactate, urine, acidosis, and tachycardia proxies	LAT_SHOCK
Respiratory failure	Hypoxemic or ventilatory failure; MIMIC can use detailed ventilation support, PhysioNet uses MechVent, FIO2, PaO2, PaCO2, SaO2, pH, and RespRate	LAT_RESPIRATORY_FAILURE
Renal dysfunction	AKI or clinically meaningful kidney dysfunction; MIMIC can add dialysis/CRRT and baseline creatinine when available, PhysioNet uses creatinine, BUN, urine, potassium, and bicarbonate	LAT_RENAL_DYSFUNCTION
Hepatic or coagulation dysfunction	Liver, platelet, or coagulation dysfunction; MIMIC can use INR/PT/PTT, while PhysioNet uses bilirubin, AST, ALT, ALP, albumin, and platelets	LAT_HEPATIC_COAG_DYSFUNCTION
Neurologic dysfunction	Depressed consciousness or acute neurologic abnormality; MIMIC can use GCS components, RASS, pupils, and sedation context, while PhysioNet uses total GCS and physiologic context	LAT_NEUROLOGIC_DYSFUNCTION
Metabolic derangement	Acid-base, lactate, glucose, or electrolyte instability; available in both, with MIMIC adding base excess/chloride and PhysioNet adding Mg and ABG variables	LAT_METABOLIC_DERANGEMENT
Cardiac strain	Myocardial injury, rhythm instability, or cardiac stress; MIMIC supports troponin, CK-MB, rhythms, and care context; PhysioNet supports Tropl/TropT, HR, BP, lactate, and ICUType	LAT_CARDIAC_STRAIN

D. Latent-Variable Determination Rules

D1. LAT_CHRONIC_BURDEN

Positive label means: high baseline vulnerability before acute ICU physiology.

Domain indicator	Rule
Age burden	AGE >= 75 in either dataset
MIMIC comorbidity burden	COMORBIDITY_COUNT >= 2 OR ELIXHAUSER_SCORE >= 5 OR Charlson-style burden elevated
MIMIC chronic organ disease	ICD evidence of chronic heart disease OR chronic lung disease OR chronic kidney disease OR chronic liver disease OR chronic neurologic disease
MIMIC malignancy or immunosuppression	ICD evidence of malignancy OR transplant OR immunosuppression
MIMIC high-risk admission context	ADMISSION_TYPE = EMERGENCY OR ADMISSION_TYPE = URGENT
PhysioNet body-reserve proxy	BMI < 18.5 OR BMI >= 40 when Height and Weight are available
PhysioNet low-albumin proxy	ALBUMIN_MIN < 3.0
PhysioNet case-mix proxy	ICUType = CCU or MICU, or analyst-specified high-risk ICU context

Final rule: For MIMIC, LAT_CHRONIC_BURDEN = 1 if at least 2 MIMIC domain indicators are positive. For PhysioNet, LAT_CHRONIC_BURDEN = 1 if at least 2 available PhysioNet proxy indicators are positive. Otherwise LAT_CHRONIC_BURDEN = 0.

Dataset-specific note: The PhysioNet version is weaker because it lacks ICD diagnoses and comorbidity scores.

D2. LAT_INFLAMMATION_SEPSIS

Positive label means: evidence of systemic inflammation, suspected infection, or sepsis-like burden.

Domain indicator	Rule
Temperature abnormality	TEMP_MAX_C >= 38.3 OR TEMP_MIN_C < 36
WBC abnormality	WBC_MAX > 12 OR WBC_MIN < 4
Lactate elevation	LACTATE_MAX > 2
Tachycardia or respiratory stress	HR_MAX > 90 OR RESP_RATE_MAX > 20 OR PACO2_MIN < 32
MIMIC culture evidence	CULTURE_ORDERED = 1 OR CULTURE_POSITIVE = 1
MIMIC antibiotic evidence	ANTIBIOTIC_STARTED = 1
PhysioNet platelet stress fallback	PLATELETS_MIN < 150 when culture/antibiotic data are unavailable

Final rule: For MIMIC, LAT_INFLAMMATION_SEPSIS = 1 if at least 2 indicators are positive AND at least one anchor is culture evidence, antibiotic evidence, temperature abnormality, or WBC abnormality. For PhysioNet, LAT_INFLAMMATION_SEPSIS = 1 if at least 3 physiologic indicators are positive OR if temperature/WBC abnormality is present with lactate elevation and at least 2 total indicators. Otherwise 0.

Dataset-specific note: PhysioNet cannot confirm infection; label should be interpreted as inflammation/sepsis-like burden rather than confirmed sepsis.

D3. LAT_GLOBAL_SEVERITY

Positive label means: multi-system acute physiological severity.

Domain indicator	Rule
Circulatory domain	MAP_MIN < 70 OR SBP_MIN <= 100 OR, in MIMIC only, VASOPRESSOR_PRESENT = 1
Respiratory domain	SPO2_OR_SAO2_MIN < 92 OR PAO2_FIO2_MIN <= 300 OR MECHANICAL_VENTILATION = 1
Neurologic domain	GCS_MIN < 15 OR, in MIMIC only, RASS_MIN <= -3 OR RASS_MAX >= 2
Renal domain	CREATININE_MAX >= 2 OR URINE_OUTPUT_24H < 500
Metabolic domain	LACTATE_MAX > 2 OR PH_MIN < 7.30 OR BICARBONATE_MIN < 18
Inflammatory domain	TEMP_MAX_C >= 38.3 OR TEMP_MIN_C < 36 OR WBC_MAX > 12 OR WBC_MIN < 4
Hepatic or coagulation domain	BILIRUBIN_MAX >= 2 OR PLATELETS_MIN < 100 OR, in MIMIC only, INR_MAX >= 1.5
Cardiac domain	TROPONIN_ABOVE_ULN = 1 OR TROP_I/TROP_T elevated OR HR_MAX >= 130

Final rule: LAT_GLOBAL_SEVERITY = 1 if at least 3 domain indicators are positive. A sensitivity version may also label positive when at least 2 domains are positive with a critical abnormality such as LACTATE_MAX >= 4, PH_MIN < 7.20, GCS_MIN <= 8, MAP_MIN < 60, or mechanical ventilation. Otherwise 0.

Dataset-specific note: This is an outcome-proximal severity marker and should be used cautiously for causal adjustment.

D4. LAT_SHOCK

Positive label means: evidence of circulatory shock or clinically meaningful hypoperfusion.

Domain indicator	Rule
Hypotension	MAP_MIN < 65 OR SBP_MIN <= 90 OR sustained MAP < 70; use invasive or non-invasive BP in PhysioNet
MIMIC vasopressor support	VASOPRESSOR_PRESENT = 1
Tissue hypoperfusion	LACTATE_MAX > 2
Oliguria	URINE_OUTPUT_6H_ML_PER_KG_HR < 0.5 OR URINE_OUTPUT_24H < 500
Acidosis	PH_MIN < 7.30 OR, in MIMIC only, BASE_EXCESS_MIN <= -5 OR, in PhysioNet fallback, BICARBONATE_MIN < 18
Tachycardic stress fallback	HR_MAX >= 110, especially in PhysioNet where vasopressors are absent

Final rule: LAT_SHOCK = 1 if at least 2 domain indicators are positive. Also assign LAT_SHOCK = 1 if MIMIC vasopressor support is present AND tissue hypoperfusion, acidosis, or hypotension is present. Otherwise LAT_SHOCK = 0.

Dataset-specific note: PhysioNet cannot identify vasopressor-supported shock, so normalized BP after treatment may create false negatives.

D5. LAT_RESPIRATORY_FAILURE

Positive label means: clinically meaningful hypoxemic or ventilatory failure.

Domain indicator	Rule
Hypoxemia	SPO2_OR_SAO2_MIN < 92 OR PAO2_MIN < 60
Oxygenation failure	PAO2_FIO2_MIN <= 300 when PaO2 and FiO2 can be paired
Respiratory support	MECHANICAL_VENTILATION = 1 OR, in MIMIC only, NONINVASIVE_VENTILATION = 1 OR, in either dataset, FIO2_MAX >= 0.5
MIMIC ventilator severity	PEEP elevated or ventilator mode indicating invasive support; absent in PhysioNet
Respiratory-rate abnormality	RESP_RATE_MAX >= 30 OR RESP_RATE_MIN <= 8; for broader severity screen, RESP_RATE_MAX >= 22
Ventilatory failure	PACO2_MAX >= 50 AND PH_MIN < 7.30

Final rule: LAT_RESPIRATORY_FAILURE = 1 if at least 2 domain indicators are positive. Also assign LAT_RESPIRATORY_FAILURE = 1 if respiratory support is present AND hypoxemia, oxygenation failure, or ventilatory failure is present. Otherwise 0.

Dataset-specific note: Elective post-operative ventilation can cause false positives, especially in CSRU/cardiac surgery contexts.

D6. LAT_RENAL_DYSFUNCTION

Positive label means: acute kidney injury or clinically meaningful renal dysfunction.

Domain indicator	Rule
Creatinine abnormality	CREATININE_MAX >= 2.0 OR CREATININE_DELTA_FROM_BASELINE >= 0.3; in PhysioNet, use within-window delta if baseline is unavailable
Azotemia	BUN_MAX >= 40
Oliguria	URINE_OUTPUT_6H_ML_PER_KG_HR < 0.5 OR URINE_OUTPUT_24H < 500
Renal metabolic consequence	POTASSIUM_MAX >= 5.5 OR BICARBONATE_MIN < 18
MIMIC renal replacement therapy	DIALYSIS_OR_CRRT_PRESENT = 1

Final rule: LAT_RENAL_DYSFUNCTION = 1 if at least 2 domain indicators are positive. Also assign LAT_RENAL_DYSFUNCTION = 1 if dialysis or CRRT is present in MIMIC. Otherwise 0.

Dataset-specific note: PhysioNet cannot distinguish chronic kidney disease from acute kidney injury without a pre-ICU baseline.

D7. LAT_HEPATIC_COAG_DYSFUNCTION

Positive label means: hepatic, platelet, or coagulation dysfunction.

Domain indicator	Rule
Bilirubin elevation	BILIRUBIN_MAX >= 2.0
Hepatocellular injury	AST_MAX >= 120 OR ALT_MAX >= 120; a stricter PhysioNet sensitivity threshold may use AST/ALT >= 200
Cholestatic injury fallback	ALP_MAX >= 250 when available, especially in PhysioNet
Thrombocytopenia	PLATELETS_MIN < 100
MIMIC coagulopathy	INR_MAX >= 1.5 OR PT_PROLONGED = 1 OR PTT_PROLONGED = 1
Low albumin	ALBUMIN_MIN < 2.5

Final rule: LAT_HEPATIC_COAG_DYSFUNCTION = 1 if at least 2 domain indicators are positive. In MIMIC, INR/PT/PTT strengthen the coagulation component. In PhysioNet, use bilirubin, AST, ALT, ALP, albumin, and platelets only. Otherwise 0.

Dataset-specific note: PhysioNet hepatic/coagulation labels are less certain because INR, PT, and PTT are absent.

D8. LAT_NEUROLOGIC_DYSFUNCTION

Positive label means: depressed consciousness or acute neurologic dysfunction.

Domain indicator	Rule
Severe GCS depression	GCS_MIN <= 8, counts as 2 points
Moderate GCS depression	GCS_MIN <= 13, counts as 1 point; GCS_MIN < 15 can be used as mild abnormality
MIMIC GCS component abnormality	abnormal eye score OR abnormal verbal score OR abnormal motor score
MIMIC RASS abnormality	RASS_MIN <= -3 OR RASS_MAX >= 2
MIMIC pupil or focal neurologic abnormality	abnormal pupil response OR focal neurologic abnormality documented
PhysioNet metabolic/gas context	Na < 130 OR Na > 150 OR Glucose < 70 OR Glucose > 300 OR PaCO2 >= 50 OR SaO2 < 90 OR pH < 7.25
Sedation or intubation context	sedative infusion present OR intubation/mechanical ventilation present; used as ambiguity flag, not positive evidence alone

Final rule: LAT_NEUROLOGIC_DYSFUNCTION = 1 if neurologic score is at least 2. If sedation or intubation is present and the neurologic score is only 1, mark as uncertain rather than positive. Otherwise LAT_NEUROLOGIC_DYSFUNCTION = 0.

Dataset-specific note: PhysioNet lacks RASS, sedation, pupil, and focal-neurology variables, so GCS-based labels may confuse sedation with neurologic injury.

D9. LAT_METABOLIC_DERANGEMENT

Positive label means: clinically meaningful acid-base, lactate, electrolyte, or glucose instability.

Domain indicator	Rule
pH abnormality	PH_MIN < 7.30 OR PH_MAX > 7.55
Bicarbonate or base abnormality	BICARBONATE_MIN < 18 OR BICARBONATE_MAX > 35 OR, in MIMIC only, BASE_EXCESS_MIN <= -5
Lactate elevation	LACTATE_MAX > 2
Potassium abnormality	POTASSIUM_MIN < 3.0 OR POTASSIUM_MAX >= 5.5
Sodium abnormality	SODIUM_MIN < 130 OR SODIUM_MAX > 150

Domain indicator	Rule
Glucose abnormality	GLUCOSE_MIN < 70 OR GLUCOSE_MAX > 250
Magnesium abnormality	MG_MIN < 0.6 OR MG_MAX > 1.2 when available
Ventilatory compensation context	PACO2_MIN < 32 OR PACO2_MAX > 50

Final rule: LAT_METABOLIC_DERANGEMENT = 1 if at least 2 domain indicators are positive. A critical-value sensitivity rule may set positive if PH_MIN < 7.20, LACTATE_MAX >= 4, or POTASSIUM_MAX >= 6.0. Otherwise 0.

Dataset-specific note: Use base excess and chloride in MIMIC when available; use HCO3, pH, lactate, electrolytes, glucose, Mg, and PaCO2 in PhysioNet.

D10. LAT_CARDIAC_STRAIN

Positive label means: evidence of myocardial injury, rhythm instability, or cardiac stress.

Domain indicator	Rule
Troponin abnormality	MIMIC: TROPONIN_ABOVE_ULN = 1 OR TROPONIN_FLAG_ABNORMAL = 1; PhysioNet: TROP_I_MAX > 0.1 OR TROP_T_MAX > 0.01, with assay-specific sensitivity checks
MIMIC CK-MB abnormality	CKMB_ABOVE_ULN = 1
Rhythm instability	MIMIC: arrhythmia documented; either dataset: HR_MAX > 150 OR HR_MIN < 40
Hemodynamic cardiac stress	HR_MAX > 130 AND hypotension present, OR HR_MIN < 40 AND hypotension present
Cardiac-care context	MIMIC FIRST_CAREUNIT = CSRU OR CCU; PhysioNet ICUType = CCU OR CSRU
Perfusion stress fallback	LACTATE_MAX > 2 with hypotension or cardiac-care context

Final rule: LAT_CARDIAC_STRAIN = 1 if at least 2 domain indicators are positive AND at least one of the following is true: troponin abnormality, CK-MB abnormality, rhythm instability, or strong cardiac-care context with hemodynamic stress. Otherwise LAT_CARDIAC_STRAIN = 0.

Dataset-specific note: PhysioNet has no rhythm diagnoses, ECG interpretation, medications, or CK-MB, so troponin missingness should be treated as unknown rather than normal.

E. Measurement and Missingness Metadata

Variable	Meaning	Recommended use
COUNT_HR	Number of heart-rate observations	Measurement intensity; both datasets
COUNT_BP	Number of blood-pressure observations	Measurement intensity; separate invasive and non-invasive BP in PhysioNet
COUNT_SPO2_SAO2	Number of oxygen-saturation observations	Respiratory monitoring intensity; SpO2 in MIMIC, SaO2 in PhysioNet
COUNT_GCS	Number of GCS observations	Neurologic measurement availability
COUNT_LACTATE	Number of lactate measurements	Shock or sepsis monitoring intensity
COUNT_ABG	Number of arterial blood gases	Respiratory and metabolic monitoring intensity
COUNT_CREATININE	Number of creatinine measurements	Renal monitoring intensity
COUNT_CBC	Number of CBC measurements	Infection or hematologic monitoring intensity
COUNT_TROPONIN	Number of troponin measurements	Cardiac suspicion and monitoring intensity
MISSING_LACTATE	Lactate unavailable in window	Do not treat as normal
MISSING_ABG	ABG unavailable in window	Use SpO2/SaO2 and FiO2 fallback when possible
MISSING_URINE_OUTPUT	Urine output unavailable in window	Do not treat as normal unless output documentation confirms no output charting
MISSING_TROPONIN	Troponin unavailable in window	Cardiac strain unknown, not absent
MISSING_GCS	GCS unavailable in window	Neurologic state uncertain
MISSING_ICD	ICD diagnosis unavailable	MIMIC chronic burden uncertain; always unavailable in PhysioNet
MISSING_TREATMENT_DATA	Medication/procedure variables unavailable	Important for PhysioNet; vasopressors, antibiotics, dialysis, non-invasive ventilation, and cultures are not observed
TOTAL_MEASUREMENT_COUNT	Total number of observations in the first window	Care intensity and observation process, not a biological state

Important rule: missingness should be stored separately from latent physiology. Missing values may reflect low suspicion, high severity, clinician concern, unit workflow, or incomplete documentation. They should not automatically be interpreted as normal clinical status. This rule is especially important when applying a common latent ontology across MIMIC and PhysioNet because the two datasets differ substantially in treatment, diagnosis, and measurement-process observability.